```
pip install nltk
```

```
Requirement already satisfied: nltk in /usr/local/lib/python3.13/dist-packages (3.9.1)
Requirement already satisfied: click in /usr/local/lib/python3.13/dist-packages (from nltk) (8.5.0)
Requirement already satisfied: joblib in /usr/local/lib/python3.13/dist-packages (from nltk) (1.5.3)
Requirement already satisfied: regex>=2021.8.3 in /usr/local/lib/python3.13/dist-packages (from nltk) (2025.
Requirement already satisfied: tqdm in /usr/local/lib/python3.13/dist-packages (from nltk) (4.67.3)
```

```
import nltk
nltk.download('wordnet')
```

```
[nltk_data] Downloading package wordnet to /root/nltk_data...
True
```

```
from nltk.stem import PorterStemmer, SnowballStemmer, WordNetLemmatizer
ps, ss, lem = PorterStemmer(), SnowballStemmer('english'), WordNetLemmatizer()
words3= ['running', 'studies', 'better', 'flies', 'meeting', 'disappointing']
print(f"{'word':<15}{'Porter':<12}{'Snowball':<12}{'Lemma (v)':<12}")
for w in words3:
  print(f"{w:<15}{ps.stem(w):<12}{ss.stem(w):<12}{lem.lemmatize(w,pos='v'):<12}")
```

```
word           Porter      Snowball    Lemma (v)
running        run         run         run
studies        studi       studi       study
better         better      better      better
flies          fli         fli         fly
meeting        meet        meet        meet
disappointing  disappoint  disappoint  disappoint
```

```
import nltk

nltk.download('punkt')
nltk.download('stopwords')
import nltk, warnings
warnings.filterwarnings('ignore')
for pkg in ['punkt_tab', 'stopwords', 'wordnet', 'movie_reviews']:
```

```
  nltk.download(pkg,quiet=True)
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.
```

```
corpus=["My Name is Muskan.",
"I am a @Data, Science. Student.",
"I am currently in 5th Semester.",
"I am pursuing. my Data Science degree from K.R. Mangalam University.",
"My University is based in Sohna, Gurugram.",]
with open('corpus.txt', 'w') as f:
  f.write('\n'.join(corpus))
print('corpus.txt written:', len(corpus), 'documents')
docs=open('corpus.txt').read().splitlines()
```

```
corpus.txt written: 5 documents
```

```
from nltk.tokenize import word_tokenize, sent_tokenize
from nltk.corpus import stopwords
import string
stop= set(stopwords.words('english'))
toks= word_tokenize(' '.join(docs).lower())
clean=[t for t in toks if t not in stop and t not in string.punctuation]
print('Before:', len(toks), 'After:', len(clean), 'tokens')
print('First 15 cleaned tokens: ', clean[:15])
```

```
Before: 45 After: 19 tokens
First 15 cleaned tokens:  ['name', 'muskan', 'data', 'science', 'student', 'currently', '5th', 'semester', '
```

```
import re
def clean_text(t):
  t=re.sub(r'http\S+', '', t)
  t=re.sub(r'@\w+','',t)
```

```
  t=re.sub(r'\d+',' ',t)
  t=re.sub(r'[^A-Za-z\s]',' ',t)
  return re.sub(r'\s+', ' ',t).strip()
print('Before:', docs[1])
print('After:', clean_text(docs[1]))
```

```
Before: I am a @Data, Science. Student.
After: I am a Science Student
```

```
%matplotlib inline
from nltk import FreqDist
import matplotlib.pyplot as plt
fd=FreqDist(clean)
print('10 most common words:')
for w, cnt in fd.most_common(10):
  print(f'{w:<12}{cnt}')
print('Hapaxes (first 8):', fd.hapaxes()[:8])
plt.figure(figsize=(8,4))
fd.plot(20, title='Top-20 Word Frequencies')
```

```
10 most common words:
data        2
science     2
university  2
name        1
muskan      1
student     1
currently   1
5th         1
semester    1
pursuing    1
Hapaxes (first 8): ['name', 'muskan', 'student', 'currently', '5th', 'semester', 'pursuing', 'degree']
<Axes: title={'center': 'Top-20 Word Frequencies'}, xlabel='Samples', ylabel='Counts'>
```